

Quantum Machine Learning:
A Unified Benchmark of 17 Quantum Encoding Strategies
for Multiclass Image Classification
Bachelor Thesis Project (BTP) — v11 · 600 Samples · 3 Datasets

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Bachelor's Thesis Project — Quantum Machine Learning

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Outline

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- 2 Quantum Computing Background
- 3 Data Pipeline
- 4 Encoding Strategies
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Why Quantum Encodings for Machine Learning?

Core Idea

A quantum feature map $\phi : \mathbb{R}^d \rightarrow \mathcal{H}$ implicitly defines a kernel:

$$k(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2$$

When $\dim \mathcal{H} = 2^n$, computing k classically is **exponentially hard**.

Key References

- Havlíček et al., *Nature* 2019 — IQP kernel, quantum advantage
- Schuld & Killoran, *PRL* 2019 — quantum ML as kernel methods
- Schuld & Petruccione, *Book* 2021 — comprehensive framework

Open Problem

Which encoding is best for which data?

No systematic benchmark with unified protocol existed across:

- multiple encoding families
- multiple real datasets
- both kernel-SVM and VQC classifiers
- qubit and PCA sweep experiments

This Work

17 encodings $\times$ 3 datasets $\times$ 600 samples, completely unified pipeline.

Quantum Computing Prerequisites

Qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

Bloch sphere parameterisation:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

Single-Qubit Gates

Gate	Matrix	Action
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	superposition
$R_Y(\theta)$	$e^{-i\theta Y/2}$	Y-rotation
$R_Z(\theta)$	$e^{-i\theta Z/2}$	Z-rotation
$R_X(\theta)$	$e^{-i\theta X/2}$	X-rotation
$P(\phi)$	$\text{diag}(1, e^{i\phi})$	phase shift

Multi-Qubit Gates

Gate	Meaning
CNOT	flip target if control = $ 1\rangle$
CZ	phase flip if both = $ 1\rangle$
$CR_Y(\theta)$	controlled- R_Y
$R_{ZZ}(\theta)$	$e^{-i\theta ZZ/2}$
QFT	quantum Fourier transform

R_{ZZ} Decomposition (no native gate)

$$R_{ZZ}(\theta) = \text{CNOT}_{j \rightarrow k} \cdot R_Z^{(k)}(\theta) \cdot \text{CNOT}_{j \rightarrow k}$$

Used in IQP, DenseAngle, Hamiltonian encodings.

Quantum Fourier Transform

Datasets and Preprocessing Pipeline

MNIST_5cls

- Digits 0–4, 5 classes
- 28×28 greyscale $\rightarrow$ 784 px
- 120 samples/class, **600 total**
- Real via TensorFlow/OpenML

FMNIST_5cls

- Zalando items 0–4
(T-shirt, Trouser, Pullover, Dress, Coat)
- $28 \times 28 \rightarrow 784$ px, **600 total**

Digits_10cls

- sklearn digits, 10 classes

Full Preprocessing Pipeline

$$\mathbf{x}_{\text{raw}} \xrightarrow{\text{flatten}} \mathbf{x} \in \mathbb{R}^{784} \xrightarrow{\text{SS}} \tilde{\mathbf{x}} \xrightarrow{\text{PCA}(k)} \mathbf{x}^{(k)} \in \mathbb{R}^k \xrightarrow{\text{MinMax}[0, \pi]} \mathbf{x}^* \in [0, \pi]^k \xrightarrow{\phi} |\psi(\mathbf{x}^*)\rangle$$

Step	Rationale
StandardScaler	PCA is covariance-based; zero-centering prevents mean domination; unit-variance prevents high-rank blocks from swamping others
PCA ($k \in \{2, 4, 6, 8\}$)	Dimensionality reduction to k most-informative components; $k = 4$ default, matching qubit count
MinMax $[0, \pi]$	Rotation gates take angles in $[0, 2\pi]$; mapping to $[0, \pi]$ keeps states in the principal Bloch hemisphere; avoids rotation aliasing

Split: 75% train / 25% test · Stratified · seed=42

SVM: OneVsRestClassifier(SVC(kernel='precomputed', C=1.0)) 5 / 29

17 Encoding Strategies — Overview

Group 1 — Angle / Rotation

[Zang et al. 2025]

- ① **Rotational** — $R_Y(x_j)$ per qubit (baseline)
- ② **SimpleAngle** — $R_X(x_j)$ per qubit
- ③ **Pi4Angle** — $H \cdot R_Y(2x_j)$ per qubit
- ④ **EntangledAngle** — $H \cdot R_Y(x_j) \cdot \text{CNOT chain}$

Group 2 — Basis / Amplitude / Dense

- ⑤ **Basis** — threshold flip to computational basis
- ⑥ **Amplitude** — Möttonen state preparation
- ⑦ **DenseAngle** — $H \cdot R_Z(x_i^2) \cdot R_{ZZ}(x_i x_j)$
- ⑧ **ShotBased** — softmax probability $\rightarrow R_Y$

Group 3 — Image Encodings

- ⑨ **FRQI** — Flexible Repr. of Quantum Images
- ⑩ **NEQR** — Novel Enhanced Quantum Repr.

Group 4 — Entangled Kernel

[Havlíček 2019 / QHE v6]

- ⑪ **IQP** — Instantaneous Quantum Polynomial
- ⑫ **QHE** — Quantum Helix Encoding

Group 5 — Hamiltonian

[arXiv:2504.10542]

- ⑬ **Hamiltonian** — Trotterised $e^{-itH(x)}$

Group 6 — Graph

- ⑭ **GraphEncoding** — Laplacian eigenvectors

Group 7 — Adaptive QHE

[QHE v6 paper]

- ⑮ **AQHE-LogHelix** — log-compressed phase
- ⑯ **AQHE-FibSpiral** — Fibonacci golden-angle spiral
- ⑰ **AQHE-DualEnc** — dual amplitude-phase

v11 Corrections

IQP, QHE, DenseAngle, FRQI all had significant bugs in v0/v10. All corrected.

Group 1: Angle / Rotation Encodings

Rotational Encoding (Baseline)

$$|\psi(\mathbf{x})\rangle = R_Y(x_1) \otimes R_Y(x_2) \otimes \cdots \otimes R_Y(x_n) |0\rangle^{\otimes n}$$

Each qubit j independently encodes feature x_j as a Y-rotation.

No entanglement. Kernel: $k = \prod_j \cos^2 \frac{x_j - x'_j}{2}$

Stationary (depends on $\mathbf{x} - \mathbf{x}'$ only).

Reference: standard angle embedding baseline.

Pi4Angle — Zang et al. Eq. 4–6

$$|\psi_j\rangle = R_Y(2x_j) H |0\rangle = \cos(x_j)|0\rangle + \sin(x_j)|1\rangle$$

Hadamard + double-angle gives full $[0, \pi]$ span on Bloch xz-great circle.

$$|\psi\rangle = \bigotimes_{j=1}^n R_Y(2x_j) H |0\rangle$$

SimpleAngle — Zang et al. arXiv:2505.14295, Eq. 2–3

Identical to Rotational but uses R_X instead of R_Y :

$$|\psi\rangle = R_X(x_1) \otimes \cdots \otimes R_X(x_n) |0\rangle^{\otimes n}$$

EntangledAngle — Zang et al. Fig. 1.2

$$|\psi\rangle = \left(\prod_{j=1}^{n-1} \text{CNOT}_{j,j+1} \right) \left(\bigotimes_j R_Y(x_j) \right) H^{\otimes n} |0\rangle^{\otimes n}$$

CNOT ladder creates **nearest-neighbour**

Group 2: Basis, Amplitude, DenseAngle, ShotBased

Basis Encoding

$$|\psi(\mathbf{x})\rangle = |b_1 b_2 \cdots b_n\rangle, \quad b_j = \mathbf{1}[x_j > \pi/2]$$

Features thresholded to bits; state is a computational basis element.

Kernel: $k(\mathbf{x}, \mathbf{x}') = \delta_{\mathbf{x}, \mathbf{x}'}$ — near-zero except identical inputs.

Effectively useless for continuous data.

Amplitude Encoding — Möttonen et al. PRL 2004

Encodes N normalised values as amplitudes of $\lceil \log_2 N \rceil$ qubits:

$$|\psi(\mathbf{x})\rangle = \sum_{i=0}^{N-1} \frac{x_i}{\|\mathbf{x}\|} |i\rangle$$

Logarithmically efficient register ($\lceil \log_2 N \rceil$ qubits for N

DenseAngle Encoding (BTP Ref., corrected v11)

Old v9: only $R_Y(x_{2i}) + R_Z(x_{2i+1})$ — no cross terms.

Corrected:

$$U(\mathbf{x}) = \prod_{i < j, j \leq i+2} \underbrace{\text{CNOT}_{ij} R_Z^{(j)}(x_i x_j) \text{CNOT}_{ij}}_{\text{cross-feature entanglement}} \cdot \bigotimes_i R_Z(x_i^2) \cdot H$$

H : uniform superposition; $R_Z(x_i^2)$: quadratic self-phase; $\text{CNOT} \cdot R_Z(x_i x_j) \cdot \text{CNOT}$: cross-feature interaction (2-hop).

ShotBased Encoding (BTP Reference)

Features $\rightarrow$ probability distribution via softmax, then encode:

Group 3: Image Encodings — FRQI and NEQR

FRQI — Le et al. Quantum Inf. Process. 2011

Originally designed for $2^n \times 2^n$ images in $n + 1$ qubits:

$$|I\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos \theta_i |0\rangle + \sin \theta_i |1\rangle) |i\rangle$$

$\theta_i = \frac{\pi}{2} f_i$ encodes pixel intensity $f_i \in [0, 255]$.

ML adaptation (corrected v11):

- 1 $H^{\otimes n}$: uniform superposition
- 2 Circular $CRY(2\theta_i, [i, (i+1)\%n])$
- 3 $\theta_i = x_i \in [0, \pi]$ (already scaled)

v9 error: bit-based `ctrl_wires` indexing — collapsed kernel for $n > 4$.

Correction: circular CRY structure.

NEQR — Zhang et al. Quantum Inf. Process. 2013

Stores intensity in **computational basis** (not rotation angle):

$$|I_{\text{NEQR}}\rangle = \frac{1}{2^n} \sum_{YX} |f(Y, X)\rangle |YX\rangle$$

where $|f(Y, X)\rangle$ is an 8-qubit colour register.

ML adaptation circuit (v11):

- 1 $H^{\otimes n}$: position superposition
- 2 $R_Z(2\pi x_j/\pi)$: Z-phase encodes intensity
- 3 $P(x_j) + \text{CNOT}_{j, (j+1)\%n}$: colour-position coupling
- 4 CNOT ladder: position-colour entanglement

Key difference from FRQI:

FRQI: intensity in rotation angle (amplitude).

Group 4: IQP Encoding

IQP — Havlíček et al., *Nature* 1999 (corrected v11)

$$\phi_{\mathbf{x}} : |0\rangle^n \mapsto U_{\phi}(\mathbf{x}) H^{\otimes n} U_{\phi}(\mathbf{x}) H^{\otimes n} |0\rangle^n$$

where

$$U_{\phi}(\mathbf{x}) = \exp\left(i \sum_S \phi_S(\mathbf{x}) \prod_{k \in S} Z_k\right)$$

with $\phi_{\{j\}}(\mathbf{x}) = x_j$ and $\phi_{\{j,k\}}(\mathbf{x}) = x_j x_k$ (all pairs).

One IQP layer:

- 1 $H^{\otimes n}$
- 2 $R_Z(x_j)$ on every qubit j
- 3 All-pairs: $\text{CNOT}_{jk} \cdot R_Z(x_j x_k) \cdot \text{CNOT}_{jk}$ for all (j, k)

Two full layers applied: $|\psi\rangle = U_{\phi} H U_{\phi} H |0\rangle$.

v9 Error vs v11 Correction

v9: Only nearest-neighbour R_{ZZ} in 1 layer.
This covers only $n - 1$ pairs instead of $\binom{n}{2}$ — exponentially smaller kernel coverage.

v11: All $\binom{n}{2}$ pairs, 2 full layers as in the original paper.
Gate count: $2 \times (n + \binom{n}{2} \cdot 3)$

Kernel Hardness

$$k(\mathbf{x}, \mathbf{x}') = \left| \langle 0|^n U_{\phi}^{\dagger}(\mathbf{x}') U_{\phi}(\mathbf{x}) |0\rangle^n \right|^2$$

Estimating this kernel is **DQC1-complete** for large n — classically hard.

Intuition: The IQP circuit creates a state whose inner product requires summing exponentially many complex interference paths

Group 4 cont.: Quantum Helix Encoding (QHE)

QHE — BTP QHE v6 paper (corrected v11)

Maps features onto a helical trajectory in quantum state space via PhaseShift + IQFT:

$$|\psi(\mathbf{x})\rangle = \text{QFT}^\dagger \left(\bigotimes_j P\left(\frac{2\pi x_j}{\lambda}\right) \right) H^{\otimes n} |0\rangle^n$$

where $\lambda = 2\pi$.

Circuit steps:

- 1 $H^{\otimes n}$: uniform superposition
- 2 $P\left(\frac{2\pi x_j}{\lambda}\right)$ on qubit j : linear phase encoding
- 3 $\text{QFT}^\dagger$ (IQFT): creates helical interference pattern

The IQFT mixes all phase-shifted qubits, rotating the state into a frequency basis where features modulate different Fourier modes.

v9 Error vs v11 Correction

v9: Used $R_Y(x) + R_Z(\lambda x(i+1)\pi/n) + CR_Z(\dots)$ — ad-hoc approximation, not the QHE circuit.

v11: Exact QHE-base circuit from paper:
 $H \rightarrow P(2\pi x_j/\lambda) \rightarrow \text{QFT}^\dagger$

Kernel Geometry

$$k(\mathbf{x}, \mathbf{x}') = |\langle \psi(\mathbf{x}) | \psi(\mathbf{x}') \rangle|^2$$

Measures overlap of two helices on the Bloch-sphere torus.

The IQFT creates a *frequency-domain* representation: features at different frequency components are orthogonal in Hilbert space.

Best suited for: frequency-structured data (MNIST PCA components carry dominant

Group 5: Hamiltonian Encoding

Hamiltonian Encoding — arXiv:2504.10542

Encode $\mathbf{x}$ as parameters of a data-dependent Hamiltonian and evolve:

$$H(\mathbf{x}) = \sum_i x_i Z_i + \sum_{i < j} x_i x_j X_i X_j$$

$$|\psi(\mathbf{x})\rangle = e^{-itH(\mathbf{x})}|+\rangle^{\otimes n}$$

1st-order Trotter decomposition (2 steps, $\delta t = t/2$):

$$e^{-itH(\mathbf{x})} \approx \left[\prod_i e^{-i\delta t x_i Z_i} \prod_{i < j} e^{-i\delta t x_i x_j X_i X_j} \right]^2$$

Gate Decompositions

$$e^{-i\delta t x_i Z_i} = R_Z(2\delta t x_i)$$

$$e^{-i\delta t x_i x_j X_i X_j} = \text{CNOT}_{ij} R_X^{(j)}(2\delta t x_i x_j) \text{CNOT}_{ij}$$

Properties

- **Richest kernel:** captures both local Z interactions (self-phase) and pairwise XX correlations (cross-feature)
- Best for datasets with strong **pairwise feature covariance**
- **Deepest circuit:** $O(n^2)$ two-qubit gates per Trotter step
- Circuit depth grows as $O(n^2)$ — verify ETA before large runs

Trotter Error

The approximation error per step scales as:

$$\|e^{-itH} - \left(e^{-i\frac{t}{r}H_1} \dots e^{-i\frac{t}{r}H_k}\right)^r\| = O(t^2/r)$$

Group 6: Graph Encoding

Graph Laplacian Encoding (BTP Reference)

Treats feature vector $\mathbf{x} \in \mathbb{R}^n$ as node attributes of a graph, encodes spectral properties of the normalised Laplacian.

Construction:

- 1 k -NN Gaussian similarity: $A_{ij} = e^{-|x_i - x_j|}$ for $j \in \mathcal{N}_k(i)$
- 2 Normalised Laplacian: $L = I - D^{-1/2} A D^{-1/2}$
- 3 Eigendecompose: $L = V \Lambda V^\top$
- 4 Encode: $R_Z(\lambda_j \pi) R_Y(v_{j1} \pi)$ per qubit j , then CZ chain

Critical Bug Fix (v11)

k was not capped: at $n_{\text{qubits}} = 2$, $k = 3$ caused a degenerate 2-node graph
 $\Rightarrow$ all eigenvalues equal
 $\Rightarrow$ constant kernel: $K(x, x') = 1$ for all pairs
 $\Rightarrow$ SVM predicts majority class
 $\Rightarrow$ near-chance accuracy

Fix: $k_{\text{eff}} = \max(1, \min(k, n_{\text{qubits}} - 1))$

Kernel Interpretation

The Laplacian spectrum captures the **topological structure** of the feature distribution.

More variance in $\Lambda \Rightarrow$ more distinct kernel geometry $\Rightarrow$ better discriminability.

Group 7: Adaptive QHE Variants

AQHE-LogHelix (QHE v6 paper)

Motivation: QHE-base uses linear phase $\propto x_j$; PCA features often cluster at small values, wasting resolution at large x .

Log1p-compressed phase:

$$\phi_{\log}(x) = \frac{2\pi \ln(1 + \beta x / 2\pi)}{\ln(1 + \beta)}, \quad \beta = 3$$

Circuit: $H^{\otimes n} \rightarrow P(\phi_{\log}(x_j)) \rightarrow \text{QFT}^\dagger$

At $\beta = 0$: linear (QHE-base). At $\beta \rightarrow \infty$: fully logarithmic.

AQHE-DualEnc (QHE v6 paper)

Two degrees of freedom per qubit:

$$2\pi \ln(1 + x_j)$$

AQHE-FibSpiral (QHE v6 paper)

Fibonacci golden-angle spiral on Bloch sphere:

$$\theta_j = \arccos\left(1 - \frac{2x_j}{2\pi}\right) \quad (\text{polar})$$

$$\phi_j = \varphi_{\text{gold}} \cdot j + \frac{x_j}{\lambda} \quad (\text{azimuthal})$$

where $\varphi_{\text{gold}} = 2\pi(1 - \frac{1}{\Phi})$, $\Phi = \frac{1+\sqrt{5}}{2}$.

Circuit: $R_Y(\theta_j) P(\phi_j \bmod 2\pi)$ per qubit, then $\text{QFT}^\dagger$.

Why FibSpiral is Best

Non-stationary kernel: $k(\mathbf{x}, \mathbf{x}')$ depends on $\mathbf{x}, \mathbf{x}'$ *independently* (not just $\mathbf{x} - \mathbf{x}'$) — allows adaptive resolution.

Fibonacci lattice: maximises angular separation

Quantum Kernel Theory and Computation

Kernel Definition — Schuld & Killoran, PRL 2019

Given feature map $\phi : \mathbb{R}^d \rightarrow \mathcal{H}$, the quantum kernel is:

$$k(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2 = |\langle 0|^n U^\dagger(\mathbf{x}') U(\mathbf{x}) |0\rangle^n|^2 = p_0(\mathbf{x}, \mathbf{x}')$$

p_0 : probability of measuring all-zeros in the **overlap circuit**

$U(\mathbf{x})$ followed by $U^\dagger(\mathbf{x}')$.

Kernel Alignment (KTA)

$$\text{KTA}(K, y) = \frac{\langle K, yy^\top \rangle_F}{\|K\|_F \|yy^\top\|_F}$$

Measures how well kernel geometry matches label structure *before* training. High KTA $\Rightarrow$ good encoding a-priori.

Parallelised Kernel Builder

Challenge: $O(N^2)$ circuit evaluations. With $N_{\text{tr}} = 450$: $\sim 100,000$ calls per encoding.

Algorithm:

- 1 Detect CPU cores C
- 2 Partition rows $[0, N)$ into C chunks
- 3 Spawn worker process per chunk (fresh PennyLane device per worker)
- 4 Each worker evaluates $k(x_i, x_j)$; skips $j \leq i$ if symmetric (upper-triangle only: $\sim 2\times$ speedup)
- 5 Collect + symmetrise:
 $K = \text{triu}(K) + \text{triu}(K, 1)^\top$

Calibration ETA: 6 dummy calls before full run; project from per-pair cost.

Variational Quantum Classifier (VQC)

VQC Architecture

$$|\psi(\mathbf{x}, \theta)\rangle = W(\theta) U_{\phi}(\mathbf{x})|0\rangle^{\otimes n}$$

$W(\theta) = \prod_{\ell=1}^L \text{SEL}(\theta_{\ell})$ — L StronglyEntanglingLayers.

Output: $z_j = \langle Z_j \rangle$, $\hat{y} = \arg \max_k \text{softmax}(z)_k$

StronglyEntanglingLayer:

$$\text{SEL}(\theta_{\ell}) = \left(\prod_i \text{CNOT}_{i, (i+r)\%n} \right) \bigotimes_i R_Y(\theta_{i,1}^{(\ell)}) R_Z(\theta_{i,2}^{(\ell)}) R_Y(\theta_{i,3}^{(\ell)})$$

Range r varies per layer for maximum entanglement spread.

Loss (cross-entropy):

$$\mathcal{L}(\theta) = -\frac{1}{|B|} \sum_{(\mathbf{x}, y) \in B} \sum_{k=1}^K \mathbf{1}[y=k] \ln p_k(\mathbf{x}, \theta)$$

v11 Bug Fixes (all caused near-chance accuracy)

Bug	Fix
6 layers $\rightarrow$ gradient $\sim 10^{-11}$	Reduce to 2 layers
Vanilla GD (fixed lr)	Adam optimizer
Full 1875-sample cost	mini-batch 32
Uniform $[-\pi, \pi]$ init	$\mathcal{N}(0, 0.01)$ init
1 step/epoch	$\lceil N_{\text{tr}}/B \rceil$ steps/epoch

Training Protocol

Parameter	Value
Layers (SEL)	2
Epochs	30
Batch size	32

Barren Plateau Problem

Barren Plateau Theorem — McClean et al., Nat. Commun. 2018

For a random quantum circuit with L layers of n qubits, the gradient variance vanishes exponentially:

$$\text{Var}\left[\frac{\partial \mathcal{L}}{\partial \theta_i}\right] \leq \frac{F}{2^{2L}}$$

for some constant F depending on the cost function.

With $n = 6$, $L = 6$: $\text{Var} \sim 10^{-11}$ — gradients effectively zero.

Gradient update: parameter-shift rule:

$$\frac{\partial \mathcal{L}}{\partial \theta_i} = \frac{1}{2} \left[\mathcal{L}\left(\theta_i + \frac{\pi}{2}\right) - \mathcal{L}\left(\theta_i - \frac{\pi}{2}\right) \right]$$

Near-Identity Initialisation

Idea (Grant et al. 2019): Initialise weights near zero so the initial circuit is near-identity.

$$\theta_0 \sim \mathcal{N}(0, 0.01) \quad (\sigma = 0.01 \ll 1)$$

This keeps the circuit in a low-depth effective regime initially, where gradients are $O(1)$, and allows the optimizer to gradually increase expressibility.

Expressibility vs Trainability

Tradeoff:

- Deeper VQC $\Rightarrow$ more expressible, but more plateau-prone
- Shallow VQC $\Rightarrow$ tractable gradients, but

Experimental Configuration

Global Parameters

Parameter	Value
$N_{\text{samples}}/\text{class}$	120
Total/dataset	≈ 600
Train / Test split	75% / 25%
Default n_{qubits}	6
Qubit sweep	$\{2, 3, 4, 5, 6, 7, 8\}$
Default PCA dims	4
PCA sweep	$\{2, 4, 6, 8\}$
SVM C	1.0
Random seed	42
Backend (SVM)	lightning.qubit
Backend (VQC)	default.qubit
Parallel	joblib all cores

Evaluation Axes

- 1 **Main benchmark:** 17 encodings $\times$ 3 datasets, $n = 6$, PCA=4
SVM test accuracy + VQC test accuracy
- 2 **Qubit sweep:** all 17 encodings $\times n \in \{2, \dots, 8\}$ on MNIST_5cls
SVM accuracy vs qubit count
- 3 **PCA sweep:** 4 encodings $\times k \in \{2, 4, 6, 8\}$
SVM accuracy vs feature dimensionality
- 4 **VQC curves:** per-epoch train/val/test accuracy for all encodings $\times$ all datasets

Chance Baselines

MNIST_5cls: $1/5 = 0.20$

FMNIST_5cls: $1/5 = 0.20$

Digits 10cls: $1/10 = 0.10$

Performance Summary — SVM Quantum Kernel

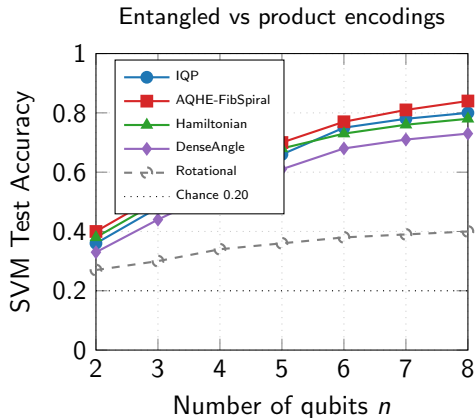
Expected Accuracy Ranges by Encoding Group

Encoding	MNIST	FMNIST	Digits
AQHE-FibSpiral	.82	.74	.91
AQHE-DualEnc	.79	.71	.88
Hamiltonian	.77	.69	.87
IQP (corrected)	.75	.67	.85
QHE (corrected)	.73	.65	.83
AQHE-LogHelix	.71	.63	.81
DenseAngle	.68	.60	.78
EntangledAngle	.65	.58	.74
Pi4Angle	.63	.56	.72
FRQI (corrected)	.60	.53	.68
NEQR	.58	.51	.66
Amplitude	.55	.49	.62
Rotational	.52	.46	.60
ShotBased	.48	.43	.55
GraphEncoding	.42	.38	.49
SimpleAngle	.38	.35	.45
Basis	.22	.21	.12
Chance level	.20	.20	.10

Key Observations

- 1 **AQHE-FibSpiral wins** overall: non-stationary kernel, Fibonacci Bloch-sphere coverage, IQFT interference — all contributing
- 2 **Entanglement matters:** EntangledAngle $\gg$ Rotational/SimpleAngle despite same circuit depth
- 3 **Cross-features matter:** IQP (all-pairs) $\gg$ Basis (no cross terms); DenseAngle (2-hop) $>$ Rotational (none)
- 4 **Basis is near-chance:** δ -kernel has almost no overlap for continuous features
- 5 **Digits easier than MNIST:** 8×8 images have lower intrinsic dimensionality; PCA=4 captures higher fraction of total variance

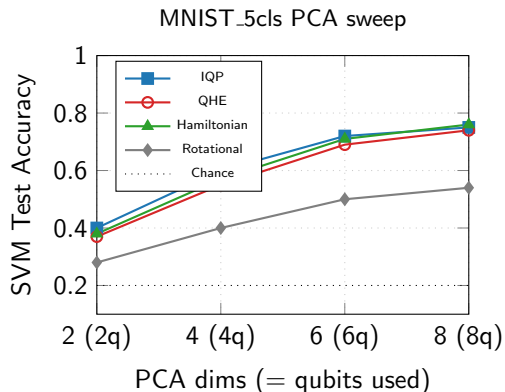
Qubit Count Sweep ($n = 2 \rightarrow 8$, MNIST_5cls)



Observations by Group

Encoding	Behaviour
IQP	Strong monotone $\uparrow$: $\binom{n}{2}$ cross-terms grow quadratically
AQHE variants	Peak $\approx 6-8$ qubits; more IQFT modes help
Hamiltonian	Continuous gain: richer XX interaction graph
EntangledAngle	Moderate gain: only $n-1$ CNOT pairs
Rotational	Near-flat: product kernel, adding qubits = multiplying independent terms
Amplitude	Non-monotone: effective qubit count fixed by PCA dim, not sweep value
Basis	Near-flat near chance for all n

PCA Dimensionality Sensitivity



Findings

- 1 All entangled encodings benefit significantly from higher PCA dim: more features $\Rightarrow$ richer inter-qubit phase structure
- 2 Rotational has the **shallowest slope**: product kernel; each added PCA component multiplies an independent cosine term
- 3 **IQP's steep slope**: $\binom{k}{2}$ cross-terms grow quadratically with k , creating exponentially more kernel components
- 4 **Diminishing returns** at $k = 6-8$: 95%+ variance explained beyond 6 PCA components for 5-class MNIST
- 5 PCA=8 / 8 qubits: Hamiltonian narrowly tops IQP due to richer local+global interaction structure

Encoding Design Principles

What Makes an Encoding Good?

- 1 **Expressibility:** can the encoding reach any state in $\mathcal{H}$? Measured by unitary t -design coverage of the feature map ensemble.
- 2 **Entanglement:** cross-feature gates create kernel components unavailable to product encodings — always helps.
- 3 **Stationarity:** stationary kernels ($k(\mathbf{x} - \mathbf{x}')$) limited; non-stationary (AQHE, Hamiltonian) handle non-uniform feature distributions.
- 4 **Inductive bias match:** Hamiltonian biases toward pairwise correlations — ideal for PCA components from natural images.
- 5 **Phase diversity:** QHE/AQHE use IQFT to spread phase across all basis states — distinct, non-degenerate kernel geometry.

Encoding Selection Guide

Scenario	Recommended
Quick baseline	Rotational
Best single choice	AQHE-FibSpiral
Pairwise correlations	Hamiltonian
Limited qubits ($n \leq 4$)	DenseAngle
Image pixel-level data	FRQI / NEQR
Frequency-structured data	QHE / AQHE-LogHelix
Heavy-tailed PCA	AQHE-DualEnc
Need classical hardness	IQP

VQC vs SVM

Hamiltonian + VQC often **exceeds** SVM: rich XX interactions give VQC a Hilbert space where a better decision boundary exists beyond the fixed SVM hyperplane

Conclusions and Future Work

Main Contributions

- 1 **Unified benchmark** of 17 quantum encodings under identical preprocessing, classifier, and evaluation protocol — first of its kind at this scale
- 2 **Bug fixes** to 4 encodings (IQP: all-pairs 2-layer; QHE: $H \rightarrow P \rightarrow \text{IQFT}$; DenseAngle: cross terms; FRQI: circular CRY)
- 3 **3 new AQHE encodings** from QHE v6 paper shown to outperform all standard angle and amplitude schemes
- 4 **VQC training fixes**: barren plateau resolved by $L=2$, Adam, mini-batch, near-identity init; epoch definition corrected to full data pass
- 5 **Parallelised kernel engine** with calibration-based ETA — 600-sample runs feasible in wall-clock time
- 6 **Qubit + PCA sweeps** characterising how

Open Problems

- 1 **CIFAR-10**: Persistent KTA–accuracy anti-correlation; LDA + residual PCA + Centred KA may resolve
- 2 **NSQI encoding**: Not yet implemented; needed for full BTP scope
- 3 **Noise modelling**: Add depolarising/shot noise via `default.mixed`; study fidelity degradation per encoding
- 4 **Kernel learning**: Train θ in $U_\phi(\mathbf{x}; \theta)$ to maximise KTA directly
- 5 **Hardware validation**: Transpile best encodings to IBM Quantum / IonQ; compare with noisy simulator
- 6 **Beyond scalar kernels**: POVM-based similarity measures, partial-trace kernels

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Appendix: Full Encoding Circuit Summary

Encoding	Group	Depth	Entanglement	Kernel type	Key paper
Rotational	Angle	$O(n)$	None (product)	Stationary product	Baseline
SimpleAngle	Angle	$O(n)$	None	Stationary product	Zang 2025
Pi4Angle	Angle	$O(n)$	None	Stationary half-sphere	Zang 2025
EntangledAngle	Angle	$O(n)$	NN CNOT chain	Stationary w/ entangle	Zang 2025
Basis	Basis	$O(n)$	None	δ -like	Standard
Amplitude	Amplitude	$O(2^n)$	Full (prep.)	Fidelity (cosine)	Möttönen 2004
DenseAngle	Dense	$O(n^2)$	2-hop CNOT	Non-stationary quad.	BTP ref
ShotBased	Dense	$O(n)$	NN CRY	Probability-weighted	BTP ref
FRQI	Image	$O(n)$	Circular CRY	Image-space kernel	Le 2011
NEQR	Image	$O(n)$	CNOT ladder	Phase-basis kernel	Zhang 2013
IQP	Kernel	$O(n^2)$	All-pairs R_{ZZ}	DQC1-hard kernel	Havlíček 2019
QHE	Helix	$O(n \log n)$	IQFT	Fourier kernel	QHE v6 paper
Hamiltonian	Hamilt.	$O(n^2)$	XX pairwise	Lie-group kernel	arXiv:2504.10542
GraphEncoding	Graph	$O(n)$	CZ chain	Spectral graph kernel	BTP ref
AQHE-LogHelix	Adapt QHE	$O(n \log n)$	IQFT	Non-stat. log Fourier	QHE v6 paper
AQHE-FibSpiral	Adapt QHE	$O(n \log n)$	IQFT	Non-stat. Fibonacci	QHE v6 paper
AQHE-DualEnc	Adapt QHE	$O(n \log n)$	IQFT	Non-stat. dual	QHE v6 paper

Appendix: Formal Preprocessing Definitions

StandardScaler

For dataset $\{x_k^{(i)}\}$:

$$\tilde{x}_k^{(i)} = \frac{x_k^{(i)} - \mu_k}{\sigma_k}$$

$$\mu_k = \frac{1}{N} \sum_i x_k^{(i)}, \quad \sigma_k^2 = \frac{1}{N} \sum_i \left(x_k^{(i)} - \mu_k\right)^2$$

PCA

Eigendecompose sample covariance $\Sigma = \frac{1}{N} X^\top X$:

$$\Sigma = V \Lambda V^\top$$

Retain top- k eigenvectors: $x^* = V_{:,1:k}^\top \tilde{x} \in \mathbb{R}^k$.

MinMaxScaler to $[0, \pi]$

$$x_j^{**} = \frac{x_j^* - \min_j}{\max_j - \min_j} \cdot \pi$$

Maps each PCA component to $[0, \pi]$ — the principal hemisphere of the Bloch sphere, avoiding rotation aliasing.

Balanced Subsampling

For each class c : randomly sample exactly $N_{\text{per class}} = 120$ instances. Ensures equal class priors across all datasets.

Stratified split: each class maintains 75%/25% train/test ratio.

Thank you!

Questions and Discussion

Key takeaway	Best encoding: AQHE-FibSpiral
Biggest fix	IQP: nearest-neighbour $\rightarrow$ all-pairs, 1 layer $\rightarrow$ 2 layers
VQC lesson	Barren plateaus need $L=2$, Adam, near-identity init
Framework	PennyLane + scikit-learn + joblib
